

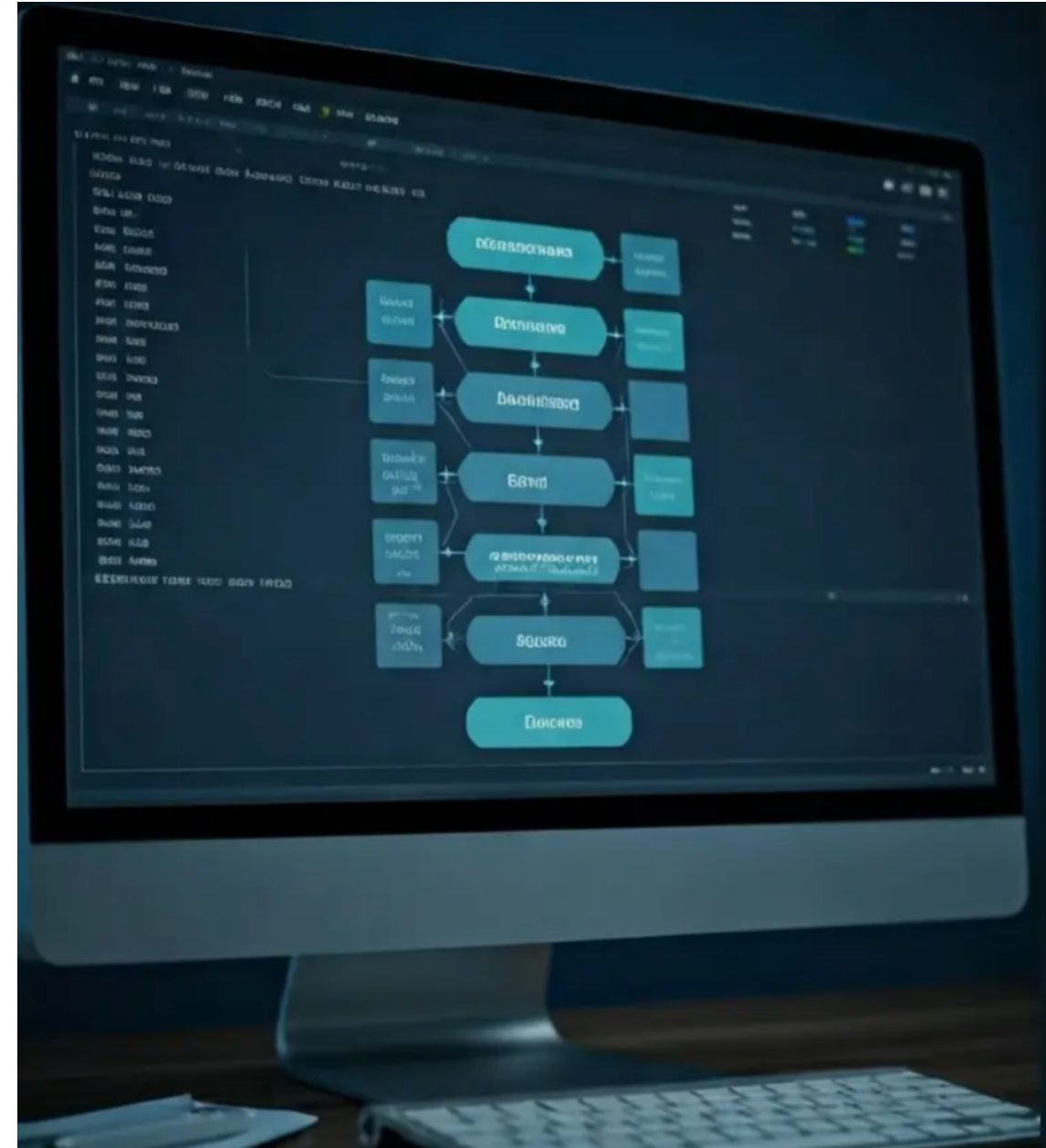
WEEK 3 – LECTURE 2

Artificial Intelligence Foundations

Informed Search Algorithms

Best-First Search • Hill Climbing • A* Search •
Heuristic Functions

CSE276 | Unit II



Quick Recap



Both BFS and DFS explore blindly.

Can AI search more intelligently instead of exploring every path?

Motivation: Making Smarter Choices



Google Maps

Suggests the *fastest* route instantly.



Robot Vacuum

Chooses the *nearest* charging station.



Delivery Drone

Avoids unnecessary detours.

How does AI know which direction looks more promising?

From Uninformed to Informed Search

Uninformed Search

- ✗ No additional knowledge
- 🔍 Explores systematically (blindly)
- ⌚ Often slow for large problems

Informed Search

- 💡 Uses clues / additional knowledge
- 🧠 Makes smarter, targeted decisions
- ⚡ Reduces unnecessary exploration

What is a Heuristic?

A heuristic is an *estimate* that helps AI decide which path appears closer to the goal.

The Google Maps Analogy

You are at point A



Map checks *Estimated Time Remaining*



Chooses the road that appears fastest



Everyday Heuristic Examples

Humans naturally use heuristics (mental shortcuts) every day.



GPS ETA

Estimating arrival based on straight-line distance.



Mall Exit

Walking towards the brightest natural light.



Parked Car

Clicking the key fob and listening for the beep.

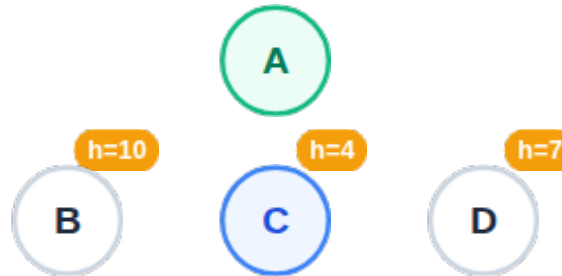


Hot and Cold

Moving in the direction where feedback gets "warmer".

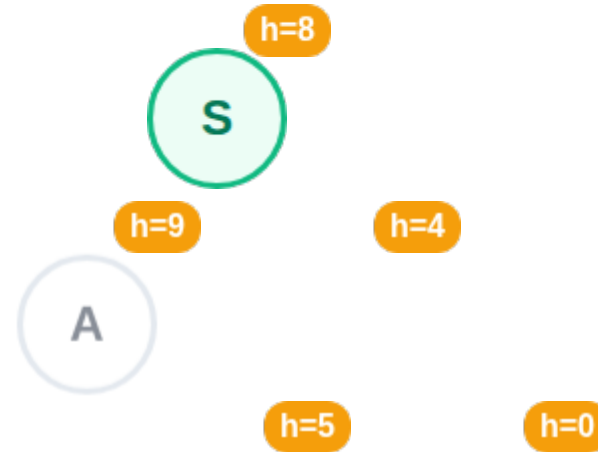
Best-First Search

Uses the heuristic value to always expand the **most promising** node first.



Best-First Search will expand 'C' because 4 is the lowest estimate.

Best-First Search Example



Start at S. Evaluate A (9) and B (4). Move to B. Evaluate C (5) and Goal (0). Move to Goal!

Characteristics of Best-First Search



Strategy

Uses heuristic estimates to guide exploration.



Efficiency

Often faster and explores fewer nodes than BFS.



Optimality

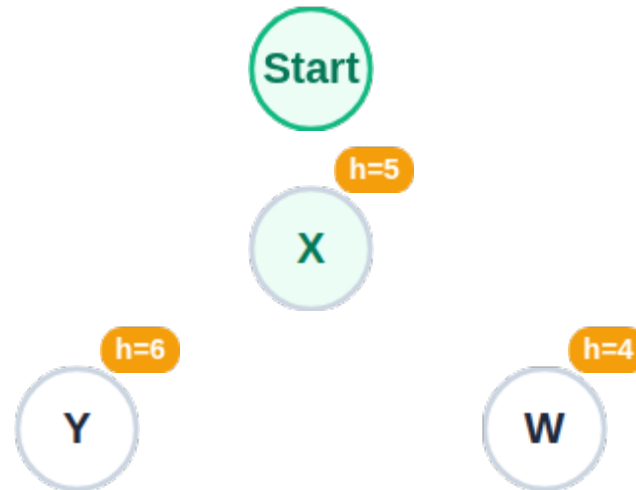
May NOT find the shortest path (can be misled by bad heuristics).



Completeness

Can get stuck in loops if not careful.

Interactive Activity



We are currently at node X.

Which node should Best-First Search explore next?

Hill Climbing

Always move to the neighbor that appears to improve the current state the most.

- ▲ **Analogy:** Climbing a mountain in thick fog.
- ↑ You only look at the steps immediately around you.
- 👣 You always take the step that goes *up* the highest.
- You stop when all neighbors are lower than your current position.



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Hill Climbing Example



Current Position



Higher Neighbor



?



Limitations of Hill Climbing



Local Maximum

A peak that is higher than its neighbors, but NOT the highest peak overall. The algorithm gets stuck.



Plateau

A flat area where all neighbors have the same value. The algorithm doesn't know which way to go.

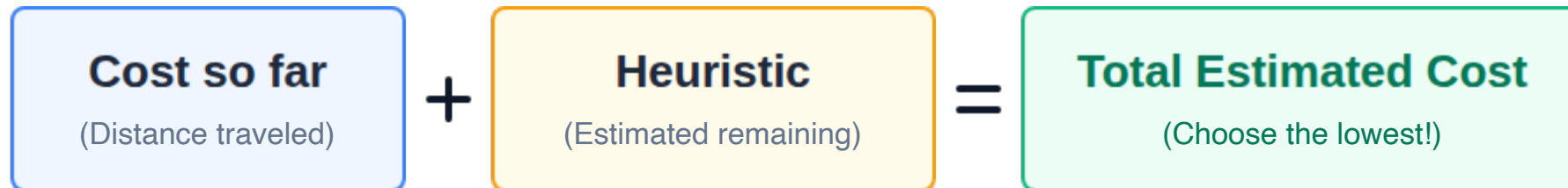


Ridge

A sequence of local maxima that is difficult for single-step moves to navigate efficiently.

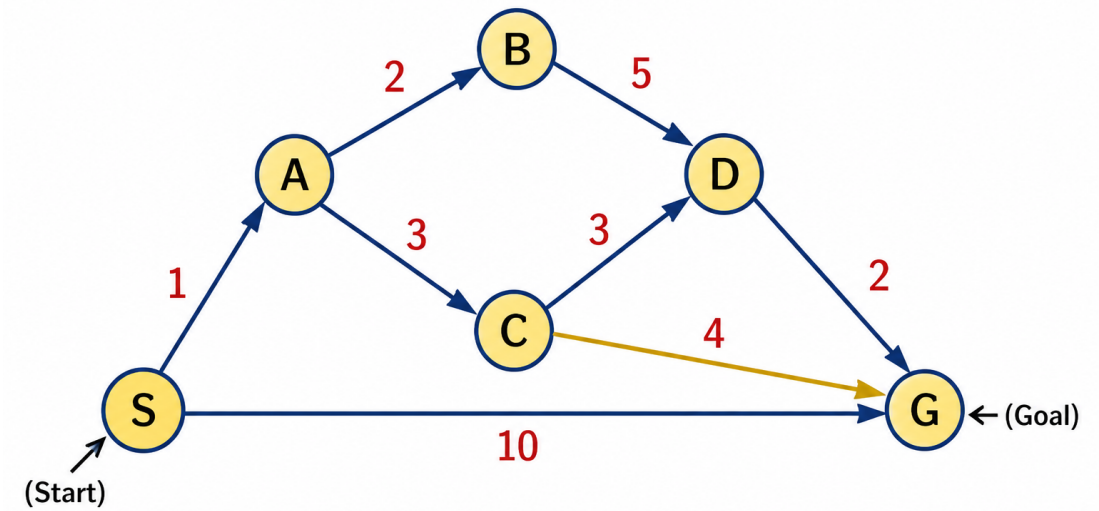
A* Search (A-Star)

The smartest of them all! A* combines the best of both worlds.



$$f(n) = g(n) + h(n)$$

A* Search Example

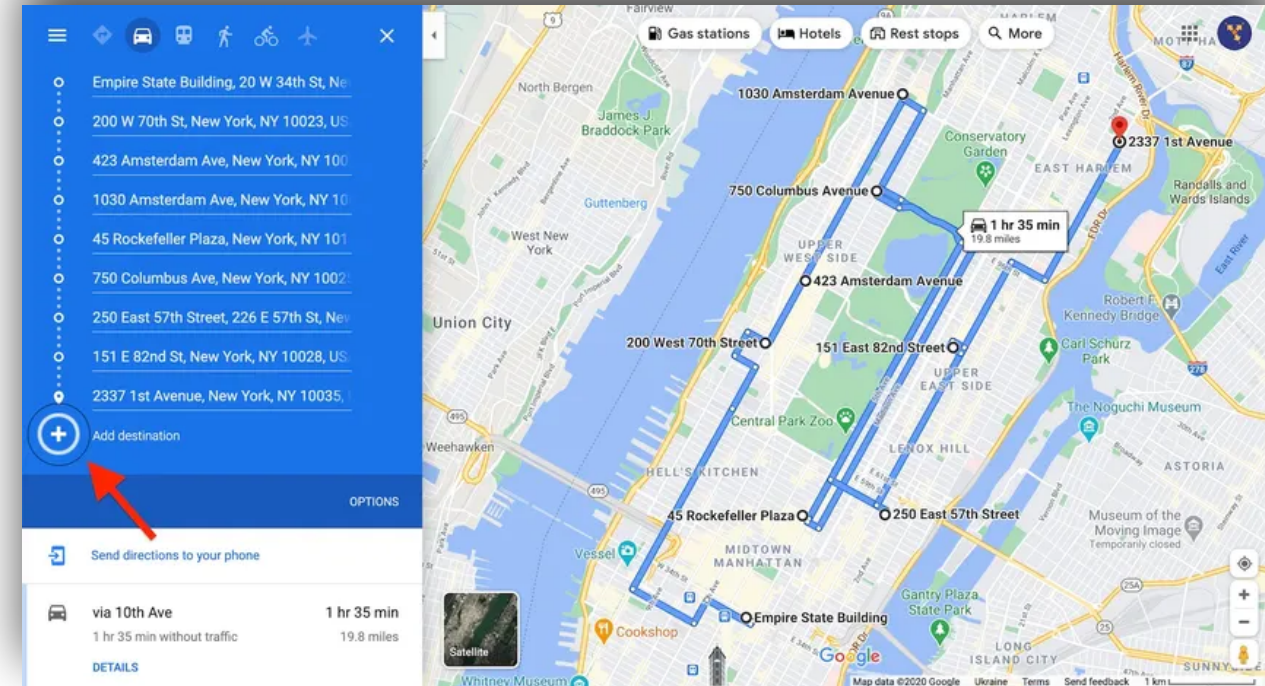
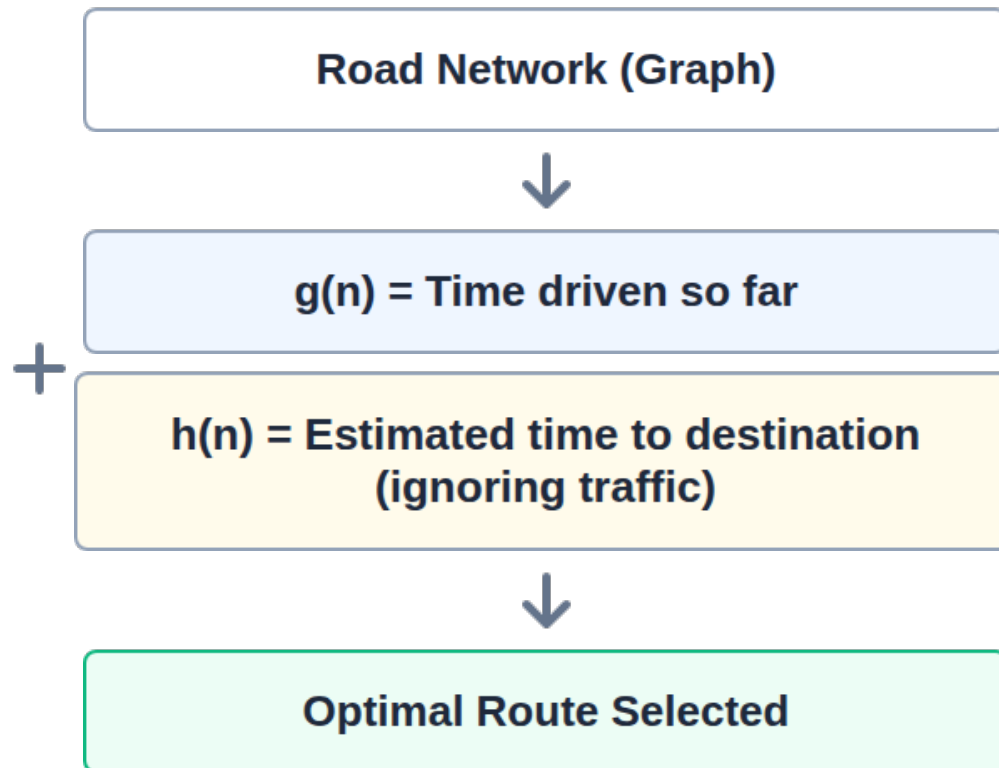


Comparing Informed Search Algorithms

Feature	 Best-First	 Hill Climbing	 A* Search
Decision Basis	Heuristic (h) only	Immediate neighbor (h)	Cost (g) + Heuristic (h)
Finds Optimal Path?	✗ No	✗ No	✓ Yes (usually)
Gets Stuck Easily?	Sometimes (loops)	✓ Yes (Local Max)	✗ No
Memory Usage	High	Very Low	Very High

Case Study: Google Maps Route Planning

Modern navigation systems use variations of A* search to navigate enormous road networks efficiently.





Board Activity

Manual Walkthrough on a Graph

-  Best-First Search path
-  Hill Climbing path
-  A* Search path

Common Misconceptions



"Heuristics always guarantee the best answer."

No, they are just estimates. A bad heuristic can lead AI astray.



"Hill Climbing never gets stuck."

It easily gets trapped in local maxima (fake peaks).



"Best-First Search and A* are identical."

Best-First ignores distance traveled so far. A* considers both traveled and estimated distance.

Key Takeaways



Informed search uses heuristics to guess the best path.



Best-First follows the most promising heuristic estimate.



Hill Climbing improves step by step but can get stuck.



A* balances cost traveled and estimated distance.

Quick Quiz

Which algorithm combines current path cost (g) with estimated remaining cost (h)?

- ☐ A) Breadth-First Search
- ☐ B) Best-First Search
- ☒ C) **A* Search**
- ☐ D) Hill Climbing

Preview: Next Lecture & Lab

Upcoming Lab:

-  Implement Hill Climbing & A*.
-  Compare efficiency against BFS/DFS.

Next Lecture: Knowledge Representation

-  How does AI store information?
-  Logic and Semantic Networks.

